

Choosing Factors

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Abstract

We examine three issues about choice of factors in the five-factor model of Fama and French (FF 2015): (i) cash profitability (*CP*) versus operating profitability (*OP*) as the variable used to construct profitability factors, (ii) long – short spread factors versus excess returns on the long or short ends of the spread factors, and (iii) factors that use the small or big ends of value, profitability, and investment factors versus averages of small and big components. We rank models primarily on the max squared Sharpe ratio for model factors, $Sh^2(f)$. This metric leads to a three-way tie for best model honors. We choose among them using other common performance metrics. The ultimate winner is the spread factor model of FF (2015) with the *OP* profitability factor replaced by a *CP* factor.

Researchers typically judge competing asset pricing models by the intercepts they leave in time series regressions of excess returns for specific assets on the models' factors. Barillas and Shanken (2016) argue that there is a more general summary measure of model performance. To see their point (in our terms, not theirs), assume the goal is to identify the model that best explains expected returns on all assets – the holy grail of asset pricing – and that we judge each competing model by the maximum (max) squared Sharpe ratio that can be constructed with the intercepts from time series regressions of excess returns on all assets on the model's factors. Define Π as returns (in excess of the riskfree rate) for all assets, f as a model's factors, a as the vector of intercepts from the regressions of Π on f , and Σ as the residual covariance matrix. The max squared Sharpe ratio for the intercepts $Sh^2(a)$ is $a'\Sigma^{-1}a$. Gibbons, Ross, and Shanken (GRS 1989) show that $a'\Sigma^{-1}a$ is the difference between the max squared Sharpe ratio one can construct with the combination of f and Π and the max one can construct with just f ,

$$a'\Sigma^{-1}a = Sh^2(f, \Pi) - Sh^2(f). \quad (1)$$

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The simple but deep insight of Barillas and Shanken (2016) is that, since the set of all assets $\mathcal{I}$ includes f , the max squared Sharpe ratio for both $\mathcal{I}$ and f is the max for just $\mathcal{I}$, $Sh^2(\mathcal{I}, f) = Sh^2(\mathcal{I})$. Thus, differences across models in the max squared Sharpe ratio for the intercepts for all assets, are driven solely by differences across models in $Sh^2(f)$,

$$a'\Sigma^{-1}a = Sh^2(\mathcal{I}) - Sh^2(f). \quad (2)$$

If the goal is to minimize the max squared Sharpe ratio for the intercepts for all assets (and we ignore measurement error), the best model is the one whose factors have the highest max squared Sharpe ratio. The intercepts for any subset of left-hand-side (LHS) assets add nothing to the information in $Sh^2(f)$.

We use this insight to compare different versions of the five-factor model of Fama and French (FF, 2015). Motivated by the dividend discount model, FF (2015) add profitability and investment factors to the three-factor model of FF (1993). The time-series regression for the model is

$$R_{it} - F_t = a_i + b_i Mkt_t + s_i SMB_t + h_i HML_t + r_i RMW_{Ot} + c_i CMA_t + e_{it}. \quad (3)$$

In this equation R_{it} is the month t return on asset i , F_t is the one-month U.S. Treasury bill rate observed at the beginning of t , Mkt_t is the return on the value-weight (VW) portfolio of NYSE-AMEX-NASDAQ stocks in excess of F_t , SMB_t (small minus big) and HML_t (high minus low book-to-market equity, BE/ME) are the size and value factors of the FF (1993) three-factor model, RMW_{Ot} (robust minus weak) is an operating profitability factor, and CMA_t (conservative minus aggressive) is an investment factor.

We study three issues about the factors in (1). The first concerns the profitability factor. Dropping the time subscript, RMW_O is constructed by sorting stocks on the accruals-based operating profitability (OP) measure suggested by Novy-Marx (2013). Ball et al (2015) argue that cash profitability (CP), that is, profitability unaffected by accruals, produces a factor that better captures average returns in sorts on accruals. We ask whether replacing RMW_O by a cash profitability factor, which we call RMW_C , increases the max squared Sharpe ratio for the model's factors. Judged on $Sh^2(f)$, the clear answer is yes.

The second issue about factor choice centers on HML , CMA , and RMW_C . The three factors are simple averages of spreads between high and low average return portfolios for small and big stocks. The

small stock component of each factor has a larger average return, and all the patterns in average returns we attempt to explain are stronger for small stocks. It is thus possible that factors that equal weight small and big stock components underestimate the premiums that drive small stock average returns and overestimate the premiums for big stocks. As an alternative, we consider models that use *Mkt*, *SMB*, and just the small or big ends of *HML*, *CMA*, and *RMW_C*.

If one uses Merton's (1973) ICAPM as motivation for multifactor models, the natural explanatory returns are the excess market return and either excess returns on the K relevant state variable mimicking portfolios or (in the terminology of Fama 1996) excess returns on K multifactor minimum variance portfolios. In the five-factor model, however, the factors paired with *Mkt* are spread portfolio returns. For example, *HML* is the difference between returns on portfolios of high and low book-to-market stocks (H and L). The third factor choice issue is whether spread factors produce higher $Sh^2(f)$ than factors that are excess returns on the long or short ends of the spread factors.

The argument that leads to $Sh^2(f)$ as the metric for judging asset pricing models assumes the goal is to find the model that minimizes the max squared Sharpe ratio of the intercepts for all assets. This goal is in line with the presumption in GRS (1989) that the fit of a model depends not just on the magnitudes of the intercepts it produces but also on the precision of the intercepts and the covariances among them, captured by Σ^{-1} in $Sh^2(a) = a'\Sigma^{-1}a$. Though intuitive and attractive, the weighting scheme for the intercepts implied by Σ^{-1} is arbitrary. It is thus interesting to examine how well $Sh^2(f)$ lines up with common measures of model performance that equal weight LHS assets. The alternative metrics we examine are those in FF (2015, 2016a, 2016b), which include (i) the average absolute intercept, (ii) a ratio that measures the dispersion of intercepts relative to the dispersion of LHS average returns, and (iii) the proportion of unexplained dispersion in average returns attributable to sampling error. Do models that produce high $Sh^2(f)$ also look good on other measures of performance?

We do not expect a uniformly positive answer to this question. The argument that competing models can be evaluated using just $Sh^2(f)$ assumes the goal is to capture expected returns on all assets. The alternative metrics we consider necessarily focus on the intercepts for specific sets of left hand side

(LHS) assets, which are a small portion of all assets. Unless a model captures expected returns on all assets, the results in Roll and Ross (1994) and Kandel and Stambaugh (1995) suggest that one can construct groups of LHS portfolios that negate or even reverse a generally positive relation between $Sh^2(f)$ and other measures of performance. For the LHS assets we examine, however, there is a strong correspondence between $Sh^2(f)$ and the alternative metrics.

To keep the discussion manageable and to limit data mining problems, we consider only competing versions of the five-factor model. To enhance our ability to see the relation between $Sh^2(f)$ and other measures of performance, we want a wide spread in the quality of models. Thus, we include models we expect to perform poorly.

Our story unfolds as follows. Section 1 presents an expression for a factor's marginal contribution to $Sh^2(f)$, which thereafter helps us understand the roles of factors in different models. Section 2 presents summary statistics for the factors. Section 3 shows that in the models we study, cash profitability (*CP*) factors produce higher $Sh^2(f)$ than operating profitability (*OP*) factors. Section 4 uses $Sh^2(f)$ to evaluate competing models that use *CP* factors. The result is a tie among three models that include: (i) *Mkt*, *SMB*, and small stock spread factors, *HML_S*, *RMW_{CS}*, and *CMA_S*; (ii) *Mkt*, *SMB*, and spread factors *HML*, *RMW_C*, and *CMA* that include small and big stocks, and (iii); a surprise winner that uses *Mkt*, *S-F*, and excess returns on the short ends of small spread factors, *L_{S-F}*, *W_{CS-F}*, and *A_{S-F}*. Section 5 analyzes the weights for factors that produce $Sh^2(f)$ for different models. Section 6 shows that there are typically strong links between $Sh^2(f)$ and other measures of model performance, but the differences between $Sh^2(f)$ and the other measures are big enough to break the tie among the three models that win on $Sh^2(f)$. Section 7 concludes.

1. Marginal Contributions to Max Squared Sharpe Ratios

The GRS (1989) result in equation (1) provides a simple way to measure a factor's marginal contribution to $Sh^2(f)$, the max squared Sharpe ratio for a model's factors. Let a_i be the intercept in the spanning regression that explains factor i returns with the model's other factors and let σ_i be the standard

deviation of the regression residuals. If we interpret factor i as the single LHS return to be explained and the other factors as f , then equation (1) implies that the increase in the max squared Sharpe ratio for a model's factors when i is added to the model is,

$$a_i^2 / \sigma_i^2 = Sh^2(f, i) - Sh^2(f). \quad (4)$$

In this equation $Sh^2(f, i)$ is the max squared Sharpe ratio for the expanded model that includes f and i , and $Sh^2(f)$ is the max squared Sharpe ratio for f . Equation (4) says a factor's marginal contribution to a model's max squared Sharpe ratio depends on both the intercept in the spanning regression of the factor on the model's other factors and the standard deviation of this regression's residuals. A factor's marginal contribution is small if its expected return is explained well by its loadings on the other factors and/or its variation not explained by the other factors is large. Intuitively, a_i^2 measures factor i 's potential contribution to the model's information about expected returns, and σ_i^2 measures the dilution of this information that occurs because the other factors in a model leave variation in factor i unexplained.

2. The Candidate Factors

Our sample is NYSE, AMEX, and NASDAQ stocks that have CRSP share codes 10 or 11 and the Compustat data required for a specific sort. (A stock can be in one set of portfolios even if it does not have the accounting data necessary for other sorts. Definitions of the sort variables are in the Appendix.) The initial tests consider 18 sets of factors. All sets include the excess market return, Mkt , but they differ on how the other four factors are defined.

HML , the value factor in regression (3), is from annual (end of June) independent sorts of stocks into two size groups and three book-to-market equity (BE/ME) groups. The accounting variables for these and other sorts at the end of June of year t are for the fiscal year ending in the previous calendar year, and market cap ME in BE/ME is for the end of December of $t-1$. The breakpoints for all factor sorts use NYSE stocks. The size break is the NYSE median market cap (ME) at the end of June, and the BE/ME breakpoints are the 30th and 70th percentiles of BE/ME for NYSE stocks. The intersections of the sorts produce six portfolios, L_S , N_S , H_S , L_B , N_B , and H_B , where L , N , and H indicate growth, neutral, and value

(low 30%, middle 40%, and high 30% of BE/ME) and subscripts S and B indicate small or big. We compute monthly value-weight (VW) returns for each portfolio from July of year t to June of $t+1$. We construct value minus growth spread factors for small and big stocks, $HML_S = H_S - L_S$ and $HML_B = H_B - L_B$, and HML , the combined spread factor, is the average of the small and big spread factors. The ends of the three spread factors provide three long excess return factors ($H-F$, H_S-F , and H_B-F) and three short excess return factors ($L-F$, L_S-F , and L_B-F).

The investment factor, CMA (conservative minus aggressive), is constructed like HML except the second sort at the end of June of year t is on Inv , the rate of growth of total assets (low to high) for the fiscal year ending in the previous calendar year. The profitability factor in (3), RMW_O (robust minus weak operating profitability), is also formed like HML except the second sort is on operating profitability (net of interest expense and scaled by book equity), again for the fiscal year ending in the previous calendar year. The cash profitability factor RMW_C mimics RMW_O except the profitability sort is on cash profits (operating profits minus the effect of accruals) divided by book equity. We again decompose each of the combined spread factors, CMA , RMW_O , and RMW_C , into small and big spread factors, three long excess return factors, and three short excess return factors. The long and short excess return factors are denoted by the first and third letters of the spread factor names.

The 2x3 sorts used to construct HML , RMW_O , RMW_C , and CMA produce four size factors, SMB_{BM} , SMB_{OP} , SMB_{CP} , and SMB_{Inv} . For example, SMB_{BM} is the average of the three small stock portfolio returns minus the average of the three big stock portfolio returns from the $ME-BE/ME$ sorts. We combine the four size factors into two overall factors; SMB_O is the average of SMB_{BM} , SMB_{OP} , and SMB_{Inv} , and SMB_C is the average of SMB_{BM} , SMB_{CP} , and SMB_{Inv} . The differences between the two overall size factors are inconsequential. The average SMB_O return is 0.26% per month ($t = 2.11$), the average SMB_C return is 0.27% ($t = 2.15$), and the correlation between the two is greater than 0.999. Because the two final size factors are so similar and because, as we shall see, models for expected returns that use cash profitability factors dominate models that use operating profitability factors, we use only SMB_C and its excess return factor, S_C-F , in the tests, and they are henceforth denoted SMB and $S-F$.

Table 1 shows summary statistics for the factors. All combined spread factors have strong average returns (premiums) during the July 1963 to December 2015 (henceforth 1963-2015) sample period. The market premium is 0.50% per month ($t = 2.82$). The value premium (average *HML* return), the investment premium (average *CMA* return), and the two profitability premiums (average *RMW_O* and *RMW_C* returns) are 0.30% to 0.39% per month and 2.95 to 6.91 standard errors above zero. The correlation (not shown) between the two profitability factors is low, 0.66. This suggests that accruals are important in the variation in *RMW_O*.

The combined spread factors *HML*, *RMW_O*, *RMW_C*, and *CMA* are averages of small and big stock spreads. The average small spreads are clustered between 0.41% and 0.52% per month, the average big spreads are between 0.18% and 0.26%, and each factor's average small spread exceeds its average big spread by at least 2.11 standard errors.

Average monthly returns for the combined long excess return factors range from 0.76% (*Ro-F*, $t = 3.72$) to 0.83% (*H-F*, $t = 4.24$). These high average returns are not surprising because in effect they all include the average excess market return, which is strong during our sample period, and they get the benefit of the high average return for small stocks, whose 50% weight in the combined factors is much greater than their weight in the market portfolio. The average returns for most of the short excess return factors are below the average excess return for the market.

Presenting summary statistics for factors is a timeworn ritual in papers that test asset pricing models. The message from equation (4), however, is that if models are judged on the max squared Sharpe ratio produced by their factors, the relevant average return for measuring the marginal contribution of a factor to a model is the intercept in the spanning regression of the factor on the model's other factors, tempered by the variance of the spanning regression residuals. We explore this issue below, but first we examine how the choice of profitability factor affects each model's max squared Sharpe ratio.

3. Operating or Cash Profitability?

One of our goals is to test whether cash profitability (*CP*) factors produce more accurate five-factor descriptions of average returns than operating profitability (*OP*) factors. Table 2 addresses this question using two sets of five-factor models. The models in the two sets are identical except one set uses *OP* profitability factors and the other uses *CP* factors.

The list of models in two sets is driven by the other asset-pricing questions we address: (i) whether factors that combine small and big components capture average returns better than factors limited to small or big stocks, and (ii) whether spread factors capture average returns better than excess return factors. As noted earlier, these two questions are primarily directed at value, profitability, and investment factors. The market portfolio is the centerpiece of every asset-pricing model, and the excess market return, *Mkt*, is in all our models. *SMB* is the size factor in all spread factor models, and *S-F* (the excess return on the small stock end of *SMB*) is the size factor in all excess return models. The latter choice reflects that fact that the correlation between *Mkt* and *B-F* exceeds 0.99, so including *B-F* rather than *S-F* would deprive a model of a size factor.

Judged on the max squared Sharpe ratio produced by a model's factors, Table 2 provides a clear answer to the choice between *OP* and *CP* profitability factors. In all nine models, the *CP* factor delivers a higher $Sh^2(f)$. The differences between $Sh^2(f_C)$ and $Sh^2(f_O)$ are typically large. For example, replacing RMW_O with RMW_C in the spread factor model of equation (1) increases $Sh^2(f)$ by almost 50%, from 0.166 to 0.242. Most of the differences between $Sh^2(f_C)$ and $Sh^2(f_O)$ are also statistically large, at least for models that have relatively high $Sh^2(f)$. Table 2 shows the 5th and 95th percentiles of the distributions of $Sh^2(f_C)$ - $Sh^2(f_O)$ from 100,000 bootstrap samples of 630 months drawn randomly with replacement from the 630 months of July 1963-December 2015. For six of nine models $Sh^2(f_C)$ exceeds $Sh^2(f_O)$ in at least 95% of the simulation runs. Two of the remaining three models are among the worst on $Sh^2(f_C)$ and $Sh^2(f_O)$, and the third is middle of the pack and so not a serious contender for best model.

Table 2 also reports intercepts and their *t*-statistics from spanning regressions in which each model's profitability factor is regressed on the other four factors. The intercept's *t*-statistic is more

extreme for the *CP* factor than for the *OP* factor for all nine models. The more extreme *t*-statistics for the *CP* factor follow from the larger max squared Sharpe ratios for the factors produced by the *CP* models. Equation (4) says that a factor's marginal contribution to a model's $Sh^2(f)$ is the ratio of the squared intercept and the residual variance, a_i^2/σ_i^2 , from the spanning regression to explain its returns. Thus, factor *i*'s marginal contribution is proportional to the squared *t*-statistic of its intercept,

$$t^2(a_i) = \lambda a_i^2 / \sigma_i^2. \quad (5)$$

The positive constant λ is the inverse of the intercept's diagonal element in the spanning regression's $(X'X)^{-1}$ matrix. Since the other four factors – the spanning regression's explanatory variables – are the same in matching *OP* and *CP* models, λ is also the same and the factor with the higher squared *t*-statistic must contribute more to $Sh^2(f)$ when it is added to the other four factors.

Confirming equations (4) and (5), Table 2 also shows that the dominance of *CP* factors depends on more than just the magnitudes of the spanning regression intercepts. For example, when the models use small spread factors (second line in the table), the intercept in the spanning regression for the *OP* version of RMW_S is larger than the intercept for the *CP* version (0.59 versus 0.54). We can infer from the *t*-statistics for the intercepts that the *CP* model wins because its spanning regression has a lower residual variance.

Judged on max squared Sharpe ratios or *t*-statistics for the intercepts in spanning regressions, *CP* factors dominate matching *OP* factors. We continue to show results for the five-factor RMW_O model of FF (2015) for perspective, but drop other *OP* models from the remaining tests.

4. Assessing Five-Factor Models with Max Squared Sharpe Ratios

Table 3 reports max squared Sharpe ratios, $Sh^2(f)$, and marginal contributions of factors to the ratios for the FF (2015) spread factor model and the nine models that use cash profitability factors. The marginal contribution of a factor to a model is the increase in $Sh^2(f)$ when that factor is added to the models's other four factors. The marginal contributions are not a breakdown of $Sh^2(f)$. In fact, for all but one case in Table 3, the sum of the marginal contributions of a model's factors is greater than its $Sh^2(f)$.

Three of the models in Table 3 use value, profitability, and investment factors that include only big stocks. Judged on $Sh^2(f)$, these models perform poorly. For example, $Sh^2(f)$ for the model that uses the big stock spread factors, HML_B , RMW_{CB} , CMA_B , is 0.078, versus 0.246 for the model that uses the small stock spread factors, HML_S , RMW_{CS} , CMA_S , and 0.242 for the model that combines small and big stock spread factors. Given the poor performance of models that use big stock factors, we focus on models that use small stock factors or factors that combine small and big stocks. $Sh^2(f)$ is higher for these models, ranging from 0.160 to 0.248.

For these better performing models, the profitability factor's marginal contribution to $Sh^2(f)$ is large, from 0.100 for the model that uses small excess return factors, H_S-F , $R_{CS}-F$ and C_S-F (and has $Sh^2(f) = 0.160$), to 0.180 for the model that uses spread factors HML , RMW_C , and CMA that combine small and big stock factors (and has $Sh^2(f) = 0.242$). The profitability factor's marginal contribution is more than half the size of $Sh^2(f)$ for all models that use small stock factors or factors that combine small and big stocks. If we drop RMW_C from the model that combines small and big stock spread factors, for example, $Sh^2(f)$ falls from 0.242 to 0.062.

Table 4 reports intercepts, t -statistics for the intercepts, and residual standard errors from spanning regressions that explain each factor in a model with the other four factors. The intercepts in the spanning regressions for Mkt in the model that uses small stock spread factors and the model that combines spread factors for small and big stocks (0.90 and 1.07) are larger than the intercepts in the spanning regressions for the cash profitability factors of the two models (0.54 and 0.51). But the residual standard errors in the spanning regressions for Mkt (3.95 and 3.79) are about three times those of the spanning regressions for the profitability factors (1.38 and 1.20). As a result, the marginal contributions of Mkt to $Sh^2(f)$ in Table 3 (0.080 and 0.052) are smaller than those of the profitability factors (0.180 and 0.154). In contrast, in the two models that combine excess return factors for small and big stocks, marginal contributions to $Sh^2(f)$ are similar for Mkt and the profitability factors of the models (Table 3). This is the result of intercepts and residual standard errors whose magnitudes are similar in the spanning regressions for Mkt and the profitability factors (Table 4).

Fama and French (2015) find that *HML*'s average return is explained by its exposure to the other four factors in the five-factor model; the intercept in their regression of *HML* on *Mkt*, *SMB*, *RMW_O*, and *CMA* is -0.04% per month and only 0.47 standard errors from zero. They infer that *HML* adds little to the FF (2015) five-factor model during their 1963-2013 period. With a longer sample, *HML*'s spanning regression intercept for the FF (2015) model in Table 4 is 0.18% per month and 2.05 standard errors from zero. Judged on marginal contributions to $Sh^2(f)$, Table 3 says *HML* and its components add little to the larger set of models examined here. For example, switching the profitability factor from *RMW_O* in the FF (2015) model to *RMW_C* increases *HML*'s spanning regression intercept to 0.30% ($t = 3.49$, Table 4), but *HML*'s marginal contribution to $Sh^2(f)$ is only 0.024 (Table 3). This is the among the largest of the marginal contributions of the value factor to $Sh^2(f)$. The other weak marginal contributor to $Sh^2(f)$ in Table 3 is the investment factor. Its largest marginal contribution, in the FF (2015) model, is 0.038.

Judged on $Sh^2(f)$, there is close to a three-way tie for best model honors in Table 3. One of the winners is the spread factor model of FF (2015) but with the cash profitability spread factor, *RMW_C*, instead of the operating profitability factor, *RMW_O*. Another winner is the model that uses the small ends of the same model's spread factors. For these two models, $Sh^2(f)$ is 0.242 and 0.246. The profitability factors of the two models make by far the largest marginal contributions to $Sh^2(f)$ (0.180 and 0.154), followed by *Mkt* (0.080 and 0.052), then *SMB* (0.039 and 0.032). The third winner, a dark horse, is the model that uses *Mkt*, *S-F*, and the excess return factors *L_S-F*, *W_{CS}-F*, and *A_S-F*, where *L_S*, *W_{CS}*, and *A_S* are the short ends of the small stock spread factors, *HML_S*, *RMW_{CS}*, and *CMA_S*. For this model, $Sh^2(f)$ is 0.248. The marginal contributions of *Mkt*, *L_S-F*, and *A_S-F* to $Sh^2(f)$ are tiny for this model (0.002, 0.003, and 0.021). *W_{CS}-F* makes a larger marginal contribution (0.127), but the small stock excess return, *S-F*, is by far the most important marginal contributor to $Sh^2(f)$. Without *S-F*, $Sh^2(f)$ falls from 0.248 to 0.018.

Are the values of $Sh^2(f)$ for the three winning models statistically distinguishable from those for other models? Table 5 summarizes 100,000 bootstrap samples that address this question. Each sample is 630 months drawn randomly with replacement from the 630 months of July 1963-December 2015. In each sample $Sh^2(f)$ is computed for the ten models of Tables 3 and 4. Table 5 shows the matrix of 5th and

95th percentiles (the 90% confidence interval) of the bootstrap distributions of $Sh^2(f_{Row})-Sh^2(f_{Column})$, the difference between $Sh^2(f)$ for the row model and $Sh^2(f)$ for the column model.

In Table 5 the three winners on $Sh^2(f)$ are models 2, 3, and 9. The table confirms that they are statistically indistinguishable: in the three comparisons of model pairs, the 5th and 95th percentiles of $Sh^2(f_{Row})-Sh^2(f_{Column})$ are roughly equidistant from zero. In contrast, $Sh^2(f)$ for each winning model (2, 3, and 9) exceeds $Sh^2(f)$ for each losing model (1, 4-8, and 10) in at least 95% of simulation runs. The only exception is the comparison between model 5 (which produces the fourth highest value of $Sh^2(f)$ in Table 3) and model 9. The 5th percentile of $Sh^2(f_9)-Sh^2(f_5)$ is slightly negative, -0.007. In short, the bootstrap simulations confirm that the three winning models produce values of $Sh^2(f)$ reliably larger than those of their seven competitors.

Skipping the details, it is worth noting that bootstrap distributions of $Sh^2(f)$ are more disperse than bootstrap distributions of paired differences between values of $Sh^2(f)$ for different models. This happens because observations of $Sh^2(f)$ are correlated across models – a likely result of the fact that Mkt is common to all models and other factors or their components are shared across models.

5. Decomposing $Sh^2(f)$

Table 3 shows the marginal contributions of each of the factors of a model to $Sh^2(f)$, the max squared Sharpe ratio for its factors. A more direct perspective on how factors combine to produce $Sh^2(f)$ is in Table 6, which shows the factor weights that deliver $T_f - F$, the monthly excess return on the tangency portfolio implied by a model's $Sh^2(f)$.

Of special interest are the results for the three models that produce the largest values of $Sh^2(f)$. Consider first the model that includes $Mkt = M-F$, SMB , and the combined spread factors, HML , RMW_C , and CMA . This model produces $Sh^2(f) = 0.242$, and the excess return on the implied tangency portfolio is,

$$T_f - F = 0.12Mkt + 0.11SMB + 0.12HML + 0.54RMW_C + 0.11CMA. \quad (6)$$

Isolating T_f , we have,

$$T_f = 0.88F + 0.12M + 0.11SMB + 0.12HML + 0.54RMW_C + 0.11CMA. \quad (7)$$

Thus, T_f is a standard portfolio (sum of the weights equal to 1.0) that invests 0.88 in the riskfree asset and 0.12 in the market portfolio, plus investments in the spread factors in which long positions in the high average return long ends of the factors are financed with offsetting short positions in the low average return short ends. In line with the marginal contributions in Table 3, Table 6 says this tangency portfolio invests heavily in the portfolio with robust profitability and makes roughly equal investments about one-fifth as large in M and the longs ends of the size, value, and investment factors. The total short allocation of T_f is the sum of the positions in the short ends of the spread factors, -0.88. In words, each dollar invested in this tangency portfolio supports \$0.88 in short positions and \$1.88 in long positions. The tangency portfolio that produces $Sh^2(f) = 0.246$ for the model that combines Mkt , SMB , and small spread factors, HML_S , RMW_{CS} , and CMA_S , weights the components of its factors in much the same way, except most of the weight for the value factor moves to the investment factor (Table 6). This is also in line with the factors' marginal contributions to $Sh^2(f)$ in Table 3.

Though the weights on the long and short ends of each spread factor are locked together, spread factor models can use both ends of the spreads to generate $Sh^2(f)$. Models that use excess return factors have only the long ends or the short ends to produce $Sh^2(f)$. This apparently leads them to take more extreme positions. Consider the excess return model that uses Mkt , $S-F$, $H-F$, R_C-F , and $C-F$. This is not one of the three winning models ($Sh^2(f) = 0.195$), but it is a useful starting point. Table 6 tells us that the excess return on its tangency portfolio is,

$$T_{f-F} = -14.07Mkt - 10.14 (S-F) + 2.24 (H-F) + 17.64(R_C-F) + 5.33 (C-F). \quad (8)$$

Since the sum of the RHS portfolio weights is 1.0 and $Mkt = M-F$, we can infer that

$$T_f = -14.07M - 10.14S + 2.24H + 17.64R_C + 5.33C. \quad (9)$$

This portfolio puts by far the heaviest positive weight on the high average return robust profitability portfolio, R_C , and lesser but still heavy weights on the conservative investment portfolio, C , and the value portfolio, H . It finances these extreme positions with heavy short sales of the market portfolio, M , and the small stock portfolio, S . Table 3 shows that M , S , and R_C make roughly equal

marginal contributions to $Sh^2(f)$, but Table 6 shows that the importance of M and S comes from short sales. The total short allocations of this tangency portfolio, -24.21, dwarf the initial \$1 invested.

The excess return factors that produce a winning $Sh^2(f) = 0.248$ combine Mkt , $S-F$, and excess returns on the short ends of the small stock spread factors, L_S-F , $W_{CS}-F$, and A_S-F , to produce

$$T_f-F = -0.29Mkt + 10.08(S-F) + 0.75(L_S-F) - 6.62(W_{CS}-F) - 2.92(A_S-F), \quad (10)$$

which implies,

$$T_f = -0.29M + 10.08S + 0.75L_S - 6.62W_{CS} - 2.92A_S. \quad (11)$$

In this set of factors only $S-F$ has a high average return, 0.79 versus about 0.50 for $M-F$, L_S-F , A_S-F , and $W_{CS}-F$ (Table 1). Perhaps as a result, the tangency portfolio takes an extreme short position in W_{CS} (-6.62) and smaller short positions in A_S (-2.92) and M (-0.29) to finance an extreme long position in S (10.08) and a more moderate position in L_S (0.75). The importance of S and W_{CS} is in line with the Table 3 evidence that the other three factors make negligible marginal contributions to $Sh^2(f)$.

The leverage in $Sh^2(f)$ for our models that use excess return factors is much higher than the leverage for models that use spread factors. For the three models that produce the highest $Sh^2(f)$, the short allocation of the model that uses Mkt , $S-F$, and excess returns on the short ends of the small stock spread factors (L_S-F , $W_{CS}-F$, A_S-F) is -9.83, versus -0.89 for the model that uses Mkt , SMB , and the small stock spread factors, (HML_S , RMW_{CS} , CMA_S), and -0.88 for the model that combines Mkt , SMB and spread factors for small and big stocks (HML , RMW_C , CMA). High leverage pushes the average monthly value of T_f-F to 3.51% for the excess return model, versus 0.46% and 0.37% for the spread models. But leverage increases volatility, and the t -statistics for the means of T_f-F are similar for the three models (12.49, 12.46, and 12.34), which must be true given the similarity of their $Sh^2(f)$.

Finally, the extreme allocations to $S-F$ (10.08), $W_{CS}-F$ (-6.62), and A_S-F (-2.92) in the model that uses Mkt , $S-F$, and excess returns on the short ends of the small stock spread factors (L_S-F , $W_{CS}-F$, A_S-F) suggest that this model focuses on the low average returns on what FF (2015, 2016a, 2016b) call the lethal combination – small firms that invest a lot despite low profitability. Portfolios of such stocks are a serious general problem for the FF five-factor model (1) in FF (2015, 2016a, 2016b). Skipping the details,

we can confirm that the winning model that uses $(L_S-F, W_{CS}-F, A_S-F)$ produces much less extreme intercepts for these troublesome portfolios than its two major competitors, at the cost of more extreme intercepts for other portfolios.

6. Comparing $Sh^2(f)$ with Other Measures of Model Performance

Leaning on Barillas and Shanken (2016), we evaluate asset pricing models by the max squared Sharpe ratios, $Sh^2(f)$, that can be produced with their factors. We next examine whether models that deliver high $Sh^2(f)$ look good on other measures of performance that summarize the intercepts from time series regressions of LHS returns on model factors. There are at least two reasons they may not. First, a central assumption in our version of the Barillas and Shanken argument is that we want the model that best describes expected returns on all assets. The LHS portfolios in empirical tests are always a subset of all assets, and the best model for all assets may not be best for subsets. The second key assumption is that we evaluate competing models on the minimum of $Sh^2(a) = a'\Sigma^{-1}a$, the max squared Sharpe ratio that can be constructed with the intercepts from regressions of LHS returns on each model's factors. Many common measures of model performance weight assets equally, but the weights in $a'\Sigma^{-1}a$ are a complicated function of the variance-covariance matrix for regression residuals. In the spirit of Gibbons, Ross, and Shanken (1989), $a'\Sigma^{-1}a$ is an attractive metric for tests of asset pricing models, but it may not be best for all applications.

6.1 Left-hand-side portfolios and measures of performance

The LHS portfolios we examine and the measures of performance we compare to $Sh^2(f)$ are those in FF (2015, 2016a, 2016b). In the initial tests of the five-factor model in FF (2015), there are six sets of LHS portfolios – three 5x5 quintile sorts on *ME* and *B/M*, *OP*, or *Inv* and three 2x4x4 sorts on *ME* and pairs of *B/M*, *OP*, and *Inv*. Since we are also interested in cash profitability, we add a 5x5 sort on *ME* and *CP*, and two 2x4x4 sorts, on *ME*, *B/M*, and *CP* and *ME*, *Inv*, and *CP*. All sorts use NYSE breakpoints, and the sorts to construct 5x5 portfolios are independent. In the 2x4x4 sorts, the first sort is on *ME* (small or big, using the NYSE median *ME* as the breakpoint) and the other two are conditional on *ME*. Thus, the

portfolios are intersections of NYSE quartiles of *B/M*, *OP*, *CP*, or *Inv* for small or big stocks. Like the factors of our models, the LHS portfolios described above are formed at the end of June each year t , and the variables in the 5x5 and 2x4x4 sorts are those in the 2x3 sorts that produce the factors. (See Appendix.)

The LHS portfolios in FF (2015) are from finer versions of the sorts that produce the right hand side (RHS) factors. For a more challenging test, FF (2016a) ask the five-factor model to describe average returns for anomalies known to cause problems for the FF (1993) three-factor model. The anomalies include (i) the flat relation between market β and average return that has long plagued tests of the CAPM (e.g. Fama and McBeth 1973), (ii) the high average returns associated with share repurchases and the low returns associated with share issues (Ikenberry, Lakonishok, and Vermaelen 1995, Loughran and Ritter 1995), (iii) the low average returns of stocks with high return volatility, measured using daily returns and daily residuals from the three-factor model (Ang Hodrick, Xing, and Zhang 2006), and (iv) the low average returns of the stocks of firms with large accounting accruals (Sloan 1996).

The anomalous patterns in average returns we examine are stronger for small stocks, and our first pass sort for the anomaly portfolios assigns stocks to NYSE quintiles on *ME*. The second sort, on the anomaly variable, also assigns stocks to NYSE quintiles, except when sorting on net share issues we form seven groups, including net repurchases, zero net issues, and quintiles of positive net issues. The first pass *ME* sorts and the second pass anomaly sorts are independent, with one exception. Large stocks with highly volatile returns are rare, so to avoid thin or empty portfolios, the second pass volatility sorts are conditional on *ME* quintile.

We use this large group of LHS portfolios to examine how well $Sh^2(f)$ lines up with the *GRS* statistic of Gibbons, Ross, and Shanken (1989), and three other measures of model performance in Fama and French (2015, 2016a, 2016b). Using A to indicate an average value, the simplest of the three is $A|a_i|$, the average absolute intercept for a set of portfolios. To estimate the cross-sectional variation in average returns not captured by a model, FF (2016a) define $\bar{r}_i$ as the difference between the average return on LHS portfolio i and the average return on the VW market. The average squared intercept over the average

squared value of $\bar{r}_i$, $Aa_i^2/A\bar{r}_i^2$, is the unexplained dispersion of LHS average returns relative to their total dispersion. FF (2016a) also report estimates of the proportion of unexplained dispersion in average returns attributable to sampling error, $As^2(a_i)/Aa_i^2$, where $As^2(a_i)$ is the average of the squared sample standard errors of the intercepts and Aa_i^2 is the average squared intercept (the numerator of $Aa_i^2/A\bar{r}_i^2$). A low value of $Aa_i^2/A\bar{r}_i^2$ is good news for an asset pricing model: it says intercept dispersion is low relative to the dispersion of LHS average returns. In contrast, a low value of $As^2(a_i)/Aa_i^2$ is bad news: it says little of the dispersion of the intercepts is sampling error rather than dispersion of the true intercepts. We call $A|a_i|$, $Aa_i^2/A\bar{r}_i^2$, and $As^2(a_i)/Aa_i^2$ equal weight (EW) metrics because the averages in the statistics weight LHS portfolios equally. We complement the *GRS* statistic and the three EW metrics with $Sh^2(a)$, the max squared Sharpe ratio for the intercepts from regressions of LHS returns on a model's factors.

6.2 Correlations between $Sh^2(f)$ and other measures of performance

Table 7 reports $Sh^2(f)$ and the other five measures of performance for the ten models of Tables 3, 4, and 5. Panel A of the table summarizes time series regressions that explain monthly returns for the 395 LHS portfolios of all LHS sorts described above. Panels B and C show results for the 100 5x5 and the 160 2x4x4 portfolios formed on *ME*, *B/M*, *OP*, *CP*, and *Inv*. Panel D shows results for the 135 anomaly portfolios. In each panel models are sorted on $Sh^2(f)$ to facilitate comparisons of $Sh^2(f)$ with other measures of performance.

The negative correlations between the max squared Sharpe ratios for the factors of different models, $Sh^2(f)$, and either the *GRS* statistic or the max squared Sharpe ratio for the intercepts for a set of LHS portfolios, $Sh^2(a)$, are not surprising. The max squared Sharpe ratio for the intercepts is the difference between $Sh^2(\pi, f)$, the max squared Sharpe ratio produced by combining the LHS assets, π , and the factors of a model, f , and $Sh^2(f)$, the max squared Sharpe ratio for the factors,

$$Sh^2(a) = Sh^2(\pi, f) - Sh^2(f). \quad (12)$$

If π includes all assets $Sh^2(\pi, f) = Sh^2(\pi)$ and all the variation in $Sh^2(a)$ across models is due to $Sh^2(f)$, so $Sh^2(a)$ and $Sh^2(f)$ are perfectly negatively correlated. Less than perfect correlation arises because

for a particular set of LHS assets there is variation across models in $Sh^2(\pi, f)$ that is not linearly related to $Sh^2(f)$.

Similarly, we can write a model's GRS statistic as

$$GRS = \gamma \left(\frac{1+Sh^2(\pi, f)}{1+Sh^2(f)} - 1 \right). \quad (13)$$

The constant γ is a function of the number of observations (630), factors (five), and LHS portfolios (395, 100, 160, and 135 in Panels A to D of Table 7). It varies across the panels of Table 7, but not across models in a panel. As with $Sh^2(a)$, cross-model variation in $Sh^2(f)$ and $Sh^2(\pi, f)$ drives variation in GRS . The ratio in (13) creates complications that don't affect (12), but the main driver for a less than perfect negative correlation between GRS and $Sh^2(f)$ is again variation in $Sh^2(\pi, f)$ that is not linearly related to $Sh^2(f)$.

The negative correlations between $Sh^2(f)$ and $Sh^2(a)$ or GRS are typically strong. The anomaly sorts in Panel D of Table 7 produce the least extreme negative correlations, -0.653 for $Sh^2(a)$ and -0.935 for GRS . The remaining correlations are below -0.93 for $Sh^2(a)$ and below -0.99 for GRS . The model whose $Sh^2(a)$ and GRS are most out of line in the anomaly sorts combines Mkt and $S-F$ with excess returns on the short ends of the spread factors ($L-F$, W_C-F , $A-F$). This model is in the middle of the pack on $Sh^2(f)$, but in the tests for the anomaly portfolios, it is better than the top three models on $Sh^2(a)$ and it competes well with them on GRS . In short, the results for the LHS anomaly portfolios show that models that work best in tests on all investible LHS assets need not be best for subsets.

The LHS assets in Panel A of Table 7 include all the LHS portfolios in Panels B, C, and D. The max Sharpe ratio for the intercepts in Panel A thus gets to exploit all the cross-section variation in the regression intercepts in Panels B, C, and D and a richer residual covariance matrix. As a result, $Sh^2(a)$ is more than five times larger in Panel A than in Panels B, C, and D. Inclusion of all assets, however, makes the multiple comparisons problem implicit in $Sh^2(a)$ most extreme in Panel A. This problem is accounted for in GRS and the model-by-model GRS statistics in Panel A are not systematically larger than those in Panels B, C, and D.

Equations (12) and (13) say $Sh^2(a)$ and GRS are directly linked to $Sh^2(f)$, so their correlations with $Sh^2(f)$ are not surprising. No direct link predicts the strong correlations we observe between $Sh^2(f)$ and the three EW measures of model performance. The correlation between $Sh^2(f)$ and the average absolute intercept, $A|a_i|$, ranges from -0.850 for the 5x5 portfolios to -0.924 for the all LHS portfolios. The correlations with the unexplained dispersion of LHS average returns $Aa_i^2/\bar{A}\bar{r}_i^2$ are all below -0.9. The correlation of $Sh^2(f)$ with the relative sampling error $As^2(a_i)/Aa_i^2$ is 0.882 for the 5x5 portfolios and more than 0.91 for the other three sets of LHS portfolios.

Judged on $Sh^2(f)$, the first three models in Table 7 tie for best model honors. Judged on the EW metrics in the table, the top line dark horse model that uses the short excess return factors L_S-F , $W_{CS}-F$, and A_S-F stumbles a bit relative to its two competitors. In the Panel A tests that use all LHS portfolios, the model loses to its two main competitors on all three EW metrics. Its problems seem to trace to the 5x5 LHS portfolios in Panel B where it more clearly loses to its two competitors on the EW metrics, and to a lesser extent to the 2x4x4 portfolios in Panel C. In the Panel D results for the anomaly portfolios, the dark horse model competes well with its two competitors on all three EW metrics.

The FF (2015) model that combines spread factors for small and big stocks but uses the operating profitability factor, RMW_O , has $Sh^2(f) = 0.166$, and on this metric it is a clear loser to the model that substitutes the cash profitability factor, RMW_C , and has $Sh^2(f) = 0.242$. Table 7 suggests, however, that judged on the EW metric $A|a_i|$, there is little space between the models that use the two profitability spread factors. On the other two EW metrics, the model that uses the CP spread factor is better in the tests for the 2x4x4 LHS portfolios (Panel C of Table 7), but the competition is a toss-up in the tests for the 5x5 LHS portfolios (Panel B) and the anomaly portfolios (Panel D). In short, judged on EW metrics, the CP spread factor does not have a clear advantage over the OP factor in some common LHS sorts.

7. Conclusions

If the goal is to explain expected returns on all investible LHS assets and the objective function is to minimize the max squared Sharpe ratio for the intercepts for all assets, ranking models on the max

squared Sharpe ratio for model factors, $Sh^2(f)$, is appropriate. This leads us to the top three models in Table 7. On the $Sh^2(f)$ metric we are indifferent among them. In applications, however, we typically seek to explain the average returns on specific sets of LHS portfolios. The tests in Table 7 say that among the three models that win on $Sh^2(f)$, the model that uses Mkt , $S-F$ and excess returns on the short ends of the small spread factors (L_S-F , $W_{CS}-F$, A_S-F) sometimes performs worse on other metrics than its two main competitors when applied to specific common sets of LHS assets, and it never performs noticeably better. In contrast, the other two winners on $Sh^2(f)$ never clearly lose on any metric we consider. The factors of these two models include Mkt , SMB , and either the small stock spread factors, HML_S , RMW_{CS} , and CMA_S , or the spread factors that use both small and big stocks, HML , RMW_C , and CMA . If forced to choose between these two models, tradition leans us toward the model that combines spread factors for small and big stocks. This is the model of FF (2015) but with the cash profitability spread factor, RMW_C , in lieu of the operating profitability factor, RMW_O .

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Appendix

The data are from the Center for Research in Security Prices (CRSP) and Compustat (supplemented with hand-collected book equity data, as in Davis, Fama, and French, 2000). We form most portfolios at the end of June in each year t , but we sort firms on the volatility measures, Var and $RVar$, every month. We include only stocks with CRSP share codes of 10 or 11. Stocks are included in every sort for which they have the necessary data. Thus, a stock may be in an *ME-Inv* portfolio for year t and not an *ME-B/M* portfolio. Stocks of firms with non-positive book equity are excluded from *B/M* sorts. The sort variables are:

ME: Market cap, price times shares outstanding.

BE/ME: The ratio of book value of equity to market value of equity. Book equity in the sort for June of year t is total assets for the last fiscal yearend in calendar year $t-1$, minus liabilities, plus balance sheet deferred taxes and investment tax credit if available, minus preferred stock liquidating value if available, or redemption value if available, or carrying value, adjusted for net share issuance from the fiscal year end to the end of December of $t-1$. Market equity (market cap) is price times shares outstanding at the end of December of $t-1$, from CRSP.

OP: Operating profitability. *OP* in the sort for June of year t is measured with accounting data for the fiscal year ending in year $t-1$ and is revenues minus cost of goods sold, minus selling, general, and administrative expenses, minus interest expense all divided by book equity. We follow Ball et al (2015) and reverse Compustat's adjustment to selling, general, and administrative expenses by subtracting research and development expenses.

CP: Cash profitability. In the sort for June of year t , *CP* is *OP* minus accruals for the fiscal year ending in $t-1$. To compute cash profitability, we follow Ball et al (2015) and define accruals as the change in accounts receivable from $t-2$ to $t-1$, plus the change in prepaid expenses, minus the change in (i) accounts payable, (ii) inventory, (iii) deferred revenue, and (iv) accrued expenses.

Inv: Investment. The change in total assets from the fiscal year ending in $t-2$ to the fiscal year ending in $t-1$ divided by total assets at $t-2$.

NS: Net stock issuance, the implied growth in split-adjusted shares outstanding from the end of June in year $t-1$ to the end of June in t . *NS* is zero if CRSP's shares outstanding does not change over the twelve months. Otherwise, we compute *NS* by comparing the total growth in *ME* from June $t-1$ to June t , $ME(t)/ME(t-1)$, with the growth implied by compounding the monthly without-dividend stock returns over the same period, $\prod(1+RetX_i)$,

$$NS = \frac{ME(t)/ME(t-1)}{\prod(1+RetX_i)} - 1.$$

Ac/B: Accruals, the change in operating working capital per split-adjusted share from $t-2$ to $t-1$ divided by book equity per split-adjusted share at $t-1$. Operating working capital is current assets minus cash and short-term investments minus current liabilities plus debt in current liabilities. We use operating working capital per split-adjusted share to adjust for the effect of changes in the scale of the firm caused by share issuances and repurchases.

β : Market β is measured at the end of June of year t . It is the sum of the current and previous months' slopes and is estimated using the preceding 60 months (24 minimum) of returns.

Var: Variance of daily total returns. Each stock's *Var* is estimated monthly using 60 days (20 minimum) of lagged returns.

RVar: Variance of daily residuals from the FF three-factor model. Each stock's *RVar* is estimated monthly using 60 days (20 minimum) of lagged returns.

Table 1 – Summary statistics for monthly factor returns, July 1963 – December 2015

Mkt is the difference between the value-weight (VW) market return, *M*, and the one-month U.S. Treasury bill rate, *F*. At the end of each June, NYSE, AMEX, and NASDAQ stocks are allocated to two size groups (Small and Big) using the NYSE median market cap (*ME*) breakpoints. Stocks are allocated independently to three *BE/ME* groups (Low to High), using NYSE 30th and 70th percentile breakpoints. The intersections of the two sorts produce six value-weight *ME-BE/ME* portfolios. In the sort for June of year *t*, *BE* is book equity at the end of the fiscal year ending in year *t*-1 and *ME* is market cap at the end of December of year *t*-1, adjusted for changes in shares outstanding between the measurement of *BE* and the end of December. The intersections of the *ME-BE/ME* sorts produce six portfolios, *L_S*, *N_S*, *H_S*, *L_B*, *N_B*, and *H_B*, where *L*, *N*, and *H* indicate low 30%, middle 40%, and high 30% of *BE/ME* and subscripts *S* and *B* indicate small or big. We compute monthly VW returns for each portfolio from July of year *t* to June of *t*+1. We construct spread factors for small and big stocks, $HML_S = H_S - L_S$ and $HML_B = H_B - L_B$, and *HML*, the combined spread factor, is the average of the small and big spread factors. The ends of the three spread factors provide three long (*H* – *F*, *H_S* – *F*, and *H_B* – *F*) and three short (*L* – *F*, *L_S* – *F*, and *L_B* – *F*) excess return factors. The investment factor, *CMA*, is constructed like *HML* except the second sort at the end of June of year *t* is on *Inv*, the rate of growth of total assets (low to high) for the fiscal year ending in the previous calendar year. The profitability factor, *RMW_O* (robust minus weak operating profitability), is also formed like *HML* except the second sort is on operating profitability (net of interest expense and scaled by book equity), again for the fiscal year ending in the previous calendar year. The cash profitability factor *RMW_C* mimics *RMW_O* except the profitability sort is on cash profits (operating profits minus the effect of accruals) divided by book equity. (The sort variables are defined in the Appendix.) Mimicking *HML*, we decompose each of the combined spread factors, *CMA*, *RMW_O*, and *RMW_C*, into small spread and big spread factors, three long excess return factors, and three short excess return factors. The long and short excess return factors are denoted by the first and third letters of the spread factor names. The 2x3 sorts used to construct *HML*, *RMW_O*, *RMW_C*, and *CMA* produce four size factors, *SMB_{BM}*, *SMB_{OP}*, *SMB_{CP}*, and *SMB_{Inv}*. For example, *SMB_{BM}* is the average of the three small stock portfolio returns minus the average of the three big stock portfolio returns from the *ME-BE/ME* sorts. *SMB* is the average of *SMB_{BM}*, *SMB_{CP}*, and *SMB_{Inv}*. The table shows averages and standard deviations of monthly factor returns and *t*-statistics for the average returns. The first row of the table shows summary statistics for *Mkt*, *SMB*, and the excess return factors *S-F* and *B-F* constructed from the long and shorts ends of *SMB*. Each of the remaining rows shows summary statistics for value (*Val*), investment (*Inv*), operating profitability (*Prfo*), cash profitability (*Prfc*), and investment (*Inv*) factors, grouped as they are in the models.

	Average Return				Standard Deviation				<i>t</i> -statistic			
	<i>Mkt</i>	<i>SMB</i>	<i>S-F</i>	<i>B-F</i>	<i>Mkt</i>	<i>SMB</i>	<i>S-F</i>	<i>B-F</i>	<i>Mkt</i>	<i>SMB</i>	<i>S-F</i>	<i>B-F</i>
Market and size factors	0.50	0.26	0.79	0.52	4.44	3.10	5.83	4.34	2.82	2.14	3.38	3.01

Table 1 (continued)

Factors		Average Return				Standard Deviation				t-statistic			
		Val	Prfo	PrfC	Inv	Val	Prfo	PrfC	Inv	Val	Prfo	PrfC	Inv
Spread factors													
Combined	<i>HML, RMW, CMA</i>	0.34	0.33	0.39	0.30	2.86	1.68	1.41	2.01	2.95	4.91	6.91	3.78
Small	<i>HML_S, RMW_S, CMA_S</i>	0.49	0.45	0.52	0.41	3.24	1.90	1.54	1.98	3.79	5.92	8.42	5.16
Big	<i>HML_B, RMW_B, CMA_B</i>	0.18	0.21	0.26	0.20	3.08	2.22	2.15	2.70	1.49	2.39	3.01	1.83
Long excess return factors													
Combined	<i>H-F, R-F, C-F</i>	0.83	0.76	0.79	0.79	4.91	5.15	4.88	4.94	4.24	3.72	4.07	4.03
Small	<i>H_S-F, R_S-F, C_S-F</i>	0.99	0.97	0.98	0.94	5.56	6.29	5.80	6.09	4.48	3.86	4.26	3.87
Big	<i>H_B-F, R_B-F, C_B-F</i>	0.66	0.56	0.60	0.65	4.64	4.47	4.41	4.35	3.60	3.15	3.41	3.74
Short excess return factors													
Combined	<i>L-F, W-F, A-F</i>	0.49	0.44	0.40	0.49	5.48	5.27	5.46	5.64	2.25	2.07	1.85	2.18
Small	<i>L_S-F, W_S-F, A_S-F</i>	0.50	0.52	0.47	0.53	6.83	6.31	6.38	6.54	1.85	2.07	1.84	2.03
Big	<i>L_B-F, W_B-F, A_B-F</i>	0.48	0.35	0.34	0.45	4.63	4.70	5.04	5.15	2.61	1.87	1.70	2.20
Small-big premiums													
Spread	<i>HML_{S-B}, RMW_{S-B}, CMA_{S-B}</i>	0.31	0.24	0.26	0.21	2.69	2.38	2.46	2.51	2.86	2.49	2.64	2.11
Long excess	<i>H_{S-B}, R_{S-B}, C_{S-B}</i>	0.33	0.41	0.38	0.29	2.92	3.62	3.33	3.79	2.81	2.82	2.90	1.92
Short excess	<i>L_{S-B}, W_{S-B}, A_{S-B}</i>	0.02	0.17	0.13	0.08	4.00	3.58	3.59	3.39	0.13	1.19	0.88	0.58

Table 2 – Comparison of five-factor models that include operating or cash profitability factors, July 1963 – December 2015

Mkt is *M-F*, the monthly excess return on the VW market portfolio. The size spread factor, *SMB*, is the difference between small stock and big stock portfolio returns (*S* and *B*). The value spread factor, *HML* (high minus low book-to-market equity), is the average of spread factors for small and big stocks, *HML_S* and *HML_B*. Long excess returns factors (*H-F* and *H_S-F*, *H_B-F*) and short excess returns factors (*L-F*, *L_S-F* and *L_B-F*) are created from the long and short ends of *HML*, *HML_S* and *HML_B*. The investment spread factor, *CMA* (conservative minus aggressive), is also the average of small stock and big stock spread factors, *CMA_S* and *CMA_B*. The long (*C-F*, *C_S-F*, *C_B-F*) and short (*A-F*, *A_S-F*, *A_B-F*) excess return factors use the long and short ends of *CMA*, *CMA_S*, and *CMA_B*. For each model there is a competition between two profitability factors. One uses operating profitability (*OP*), and the other uses cash profitability (*CP*). The profitability factors (*RMW*, *RMW_S*, *RMW_B*, *R-F*, *R_S-F*, *R_B-F*, *W-F*, *W_S-F*, and *W_B-F*) are constructed in the same way as the value and investment factors. Each line of the table shows (i) the factors in a model, (ii) $Sh^2(f_o)$, the max squared Sharpe ratio for the model's factors when the profitability factor uses *OP*, (iii) $Sh^2(f_c)$, the max squared Sharpe ratio for the model's factors when the profitability factor uses *CP*, and (iv) the intercepts (a_i) and *t*-statistics for the intercepts for the alternative profitability factors, Prf_o and Prf_c . The table also shows the 5th and 95th percentiles (90% confidence interval) of the distributions of $Sh^2(f_c)-Sh^2(f_o)$ from 100,000 bootstrap samples of 630 months drawn randomly with replacement from the 630 months of July 1963-December 2015.

Model Factors		$Sh^2(f_o)$	$Sh^2(f_c)$	90% Interval $Sh^2(f_c)-Sh^2(f_o)$	a_i Prf_o	Prf_c	t -statistic(a_i) Prf_o	Prf_c
Spread factors								
Combined	<i>Mkt, SMB, HML, RMW, CMA</i>	0.166	0.242	0.031 : 0.126	0.49	0.51	7.87	10.33
Small	<i>Mkt, SMB, HML_S, RMW_S, CMA_S</i>	0.208	0.246	0.004 : 0.076	0.59	0.54	8.16	9.41
Big	<i>Mkt, SMB, HML_B, RMW_B, CMA_B</i>	0.065	0.078	-0.012 : 0.041	0.33	0.38	4.59	5.38
Long excess return factors								
Combined	<i>Mkt, S-F, H-F, R-F, C-F</i>	0.133	0.195	0.030 : 0.099	0.19	0.21	7.02	9.29
Small	<i>Mkt, S-F, H_S-F, R_S-F, C_S-F</i>	0.145	0.160	-0.021 : 0.054	0.24	0.23	7.10	7.70
Big	<i>Mkt, S-F, H_B-F, R_B-F, C_B-F</i>	0.068	0.089	0.006 : 0.039	0.14	0.18	4.40	5.65
Short excess return factors								
Combined	<i>Mkt, S-F, L-F, W-F, A-F</i>	0.133	0.185	0.008 : 0.103	-0.26	-0.29	-6.41	-8.48
Small	<i>Mkt, S-F, L_S-F, W_S-F, A_S-F</i>	0.177	0.248	0.039 : 0.108	-0.26	-0.27	-5.65	-8.45
Big	<i>Mkt, S-F, L_B-F, W_B-F, A_B-F</i>	0.041	0.044	-0.013 : 0.023	-0.14	-0.19	-2.80	-3.17

Table 3 – Max squared Sharpe ratios and factor marginal contributions to max squared Sharpe ratios, July 1963–December 2015

Mkt is *M-F*, the monthly excess return on the VW market portfolio. The size spread factor, *SMB*, is the difference between small stock and big stock portfolio returns (*S* and *B*). The value spread factor, *HML* (high minus low book-to-market equity), is the average of spread factors for small and big stocks, *HML_S* and *HML_B*. Long excess returns factors (*H-F* and *H_S-F*, *H_B-F*) and short excess returns factors (*L-F*, *L_S-F* and *L_B-F*) are created from the long and short ends of *HML*, *HML_S* and *HML_B*. The investment spread factor, *CMA* (conservative minus aggressive), is also the average of small stock and big stock spread factors, *CMA_S* and *CMA_B*. The long (*C-F*, *C_S-F*, *C_B-F*) and short (*A-F*, *A_S-F*, *A_B-F*) excess return factors use the long and short ends of *CMA*, *CMA_S*, and *CMA_B*. The profitability factors are constructed in the same way as the value and investment factors. The profitability factor *RMW₀* for the FF (2015) model in the first line of the table uses operating profitability. The profitability factors in the remaining lines of the table use cash profitability, indicated with a subscript C. Each line of the table shows (i) the factors in a model, (ii) $Sh^2(f)$, the max squared Sharpe ratio for the model's factors, and (iii) the marginal contributions of *Mkt* and the model's Size, value (*Val*), profitability (*Prf*), and investment (*Inv*) factors to $Sh^2(f)$. The marginal contribution of a factor to $Sh^2(f)$ is the square of the ratio of the intercept in the spanning regression of the factor on the model's other four factors (a_i) to the standard error of the regressions residuals (σ_i).

		Marginal Contribution to $Sh^2(f) = (a_i/\sigma_i)^2$					
	Model Factors	$Sh^2(f)$	<i>Mkt</i>	<i>Size</i>	<i>Val</i>	<i>Prf</i>	<i>Inv</i>
FF (2015)	<i>Mkt, SMB, HML, RMW₀, CMA</i>	0.166	0.038	0.016	0.008	0.104	0.038
Spread factors (cash profitability)							
Combined	<i>Mkt, SMB, HML, RMW_C, CMA</i>	0.242	0.080	0.039	0.024	0.180	0.009
Small	<i>Mkt, SMB, HML_S, RMW_{CS}, CMA_S</i>	0.246	0.052	0.032	0.002	0.154	0.028
Big	<i>Mkt, SMB, HML_B, RMW_{CB}, CMA_B</i>	0.078	0.033	0.006	0.010	0.047	0.005
Long excess return factors (cash profitability)							
Combined	<i>Mkt, S-F, H-F, R_C-F, C-F</i>	0.195	0.154	0.131	0.016	0.144	0.031
Small	<i>Mkt, S-F, H_S-F, R_{CS}-F, C_S-F</i>	0.160	0.000	0.113	0.031	0.100	0.018
Big	<i>Mkt, S-F, H_B-F, R_{CB}-F, C_B-F</i>	0.089	0.066	0.034	0.007	0.053	0.013
Short excess return factors (cash profitability)							
Combined	<i>Mkt, S-F, L-F, W_C-F, A-F</i>	0.185	0.114	0.161	0.010	0.121	0.003
Small	<i>Mkt, S-F, L_S-F, W_{CS}-F, A_S-F</i>	0.248	0.002	0.230	0.003	0.127	0.021
Big	<i>Mkt, S-F, L_B-F, W_{CB}-F, A_B-F</i>	0.044	0.007	0.004	0.001	0.016	0.008

Table 4 – Spanning regression intercepts, *t*-statistics, and residual standard errors, July 1963 – December 2015

Mkt is *M-F*, the monthly excess return on the VW market portfolio. The *Size* factor in the spread factor models is *SMB*, the difference between small stock and big stock portfolio returns (*S* and *B*). The *Size* factor in the models that use excess returns is *S-F*. The combined value (*Val*) spread factor, *HML* (high minus low book-to-market equity), is the average of spread factors for small and big stocks, *HML_S* and *HML_B*. Long *Val* excess returns factors (*H-F* and *H_S-F*, *H_B-F*) and short *Val* excess returns factors (*L-F*, *L_S-F* and *L_B-F*) are created from the long and short ends of *HML*, *HML_S* and *HML_B*. The combined investment (*Inv*) spread factor, *CMA* (conservative minus aggressive), is also the average of small stock and big stock *Inv* spread factors, *CMA_S* and *CMA_B*. The long (*C-F*, *C_S-F*, *C_B-F*) and short (*A-F*, *A_S-F*, *A_B-F*) excess return factors use the long and short ends of *CMA*, *CMA_S*, and *CMA_B*. The profitability (*Prf*) factors are constructed in the same way as the *Val* and *Inv* factors. The *Prf* factor for the FF (2015) spread factor model in the first line of the table uses operating profitability. The *Prf* factors in the remaining lines of the table use cash profitability. Each line of the table shows (i) the intercept, a_i , in the spanning regression of each factor in a model on the model's other four factors, (ii) the *t*-statistics for the intercepts, and (iii) the residual standard errors, σ_i , from the spanning regressions. The intercepts and residual standard errors are the components of the ratios that determine the marginal contributions of factors to the max squared Sharpe ratio produced by the factors of a model

	Intercept $\equiv a_i$					<i>t</i> -statistic(a_i)					Residual Standard Error $\equiv \sigma_i$				
	<i>Mkt</i>	<i>Size</i>	<i>Val</i>	<i>Prf</i>	<i>Inv</i>	<i>Mkt</i>	<i>Size</i>	<i>Val</i>	<i>Prf</i>	<i>Inv</i>	<i>Mkt</i>	<i>Size</i>	<i>Val</i>	<i>Prf</i>	<i>Inv</i>
FF (2105)	0.77	0.37	0.18	0.49	0.27	4.63	2.93	2.05	7.87	4.61	3.95	2.92	2.00	1.51	1.36
Spread factors (cash profitability)															
Combined	1.07	0.56	0.30	0.51	0.13	6.59	4.53	3.49	10.33	2.20	3.79	2.81	1.94	1.20	1.38
Small	0.90	0.49	0.10	0.54	0.27	5.22	4.08	0.92	9.41	3.83	3.95	2.71	2.42	1.38	1.58
Big	0.71	0.23	0.21	0.38	0.15	4.44	1.91	2.40	5.38	1.79	3.91	2.91	2.09	1.77	1.99
Long excess return factors (cash profitability)															
Combined	-0.29	-0.34	0.19	0.21	0.15	-9.66	-8.81	2.94	9.29	4.07	0.73	0.93	1.48	0.56	0.85
Small	-0.02	-0.17	0.25	0.23	0.14	-0.18	-8.24	4.15	7.70	3.11	1.98	0.52	1.43	0.73	1.02
Big	-0.14	0.43	0.15	0.18	0.15	-6.37	4.50	1.96	5.65	2.76	0.54	2.31	1.83	0.78	1.30
Short excess return factors (cash profitability)															
Combined	0.33	0.44	-0.07	-0.29	-0.04	8.18	9.96	-2.28	-8.48	-1.27	0.97	1.10	0.74	0.83	0.66
Small	-0.08	0.33	0.06	-0.27	-0.10	-0.89	11.94	1.26	-8.45	-3.32	1.97	0.68	1.06	0.77	0.72
Big	0.05	0.16	0.03	-0.19	-0.09	2.05	1.63	0.82	-3.17	-2.18	0.59	2.41	0.81	1.45	1.04

Table 5 – 90% confidence intervals of the distributions of $Sh^2(f_{Row})-Sh^2(f_{Column})$ from 100,000 bootstrap simulations.

Each bootstrap sample is 630 months drawn randomly with replacement from the 630 months of July 1963-December 2015. In each bootstrap sample $Sh^2(f)$ is computed for the ten models of Panel A. Panel B shows the matrix of the 5th and 95th percentiles (the 90% confidence interval) of the bootstrap distributions of $Sh^2(f_{Row})-Sh^2(f_{Column})$, the difference between $Sh^2(f)$ for the row number model and $Sh^2(f)$ for the column number model. When the 5th and 95th percentiles of the bootstrap distribution of $Sh^2(f_{Row})-Sh^2(f_{Column})$ are positive, $Sh^2(f_{Row})$ is greater than $Sh^2(f_{Column})$ is at least 90% of simulation runs, and we conclude that the 1963-2015 actual $Sh^2(f)$ for the row model is reliably higher than the 1963-2015 actual $Sh^2(f)$ for the column model. When the 5th and 95th percentiles of the bootstrap distribution of $Sh^2(f_{Row})-Sh^2(f_{Column})$ are negative, $Sh^2(f_{Row})$ is less than $Sh^2(f_{Column})$ is at least 90% of simulation runs, and we conclude that the 1963-2015 actual $Sh^2(f)$ for the row model is reliably less than $Sh^2(f)$ for the column model. When the two models are pairs of models 2, 3, and 9 (the three winning models on $Sh^2(f)$ in previous tables, the 5th and 95th percentiles of the bootstrap distribution of $Sh^2(f_{Row})-Sh^2(f_{Column})$ are roughly equidistant from zero, which means the 1963-2015 $Sh^2(f)$ for the two models are statistically indistinguishable.

Panel A: Models and their identifying numbers

1	FF (2015)	<i>Mkt, SMB, HML, RMW_O, CMA</i>
2	Combined spread factors	<i>Mkt, SMB, HML, RMW_C, CMA</i>
3	Small spread factors	<i>Mkt, SMB, HML_S, RMW_{CS}, CMA_S</i>
4	Big spread factors	<i>Mkt, SMB, HML_B, RMW_{CB}, CMA_B</i>
5	Combined long excess return factors	<i>Mkt, S-F, H-F, R_C-F, C-F</i>
6	Small long excess return factors	<i>Mkt, S-F, H_S-F, R_{CS}-F, C_S-F</i>
7	Big long excess return factors	<i>Mkt, S-F, H_B-F, R_{CB}-F, C_B-F</i>
8	Combined short excess return factors	<i>Mkt, S-F, L-F, W_C-F, A-F</i>
9	Small short excess return factors	<i>Mkt, S-F, L_S-F, W_{CS}-F, A_S-F</i>
10	Big short excess return factors	<i>Mkt, S-F, L_B-F, W_{CB}-F, A_B-F</i>

Panel B: 90% confidence interval for distributions of $Sh^2(f_{Row})-Sh^2(f_{Column})$

	1	2	3	4	5	6	7	8	9
2	0.031: 0.126								
3	0.029: 0.136	-0.048: 0.057							
4	-0.133: -0.047	-0.211: -0.125	-0.238: 0.106						
5	-0.013: 0.072	-0.094: -0.004	-0.106: 0.001	0.072: 0.167					
6	-0.058: 0.046	-0.141: -0.028	-0.119: 0.058	0.026: 0.142	-0.077: 0.007				
7	-0.120: -0.039	-0.211: -0.103	-0.229: 0.094	-0.018: 0.039	-0.145: 0.073	-0.131: 0.016			
8	-0.030: 0.071	-0.086: -0.030	-0.120: 0.005	0.073: 0.148	-0.069: 0.052	-0.036: 0.089	0.043: 0.156		
9	0.033: 0.134	-0.050: 0.061	-0.029: 0.032	0.109: 0.237	-0.007: 0.117	0.037: 0.142	0.096: 0.228	0.009: 0.118	
10	-0.170: 0.080	-0.256: 0.149	-0.271: 0.143	-0.057: 0.015	-0.207: 0.100	-0.175: 0.063	-0.081: 0.011	-0.189: 0.100	-0.271: 0.145

Table 6 – Properties of tangency portfolio that maximizes the squared Sharpe ratio for a model's factors, July 1963 – December 2015

Mkt is *M-F*, the monthly excess return on the VW market portfolio. The *Size* spread factor, *SMB*, is the difference between small stock and big stock portfolio returns (*S* and *B*). The value (*Val*) spread factor, *HML* (high minus low book-to-market equity), is the average of spread factors for small and big stocks, *HML_S* and *HML_B*. Long excess returns factors (*H-F* and *H_S-F*, *H_B-F*) and short excess returns factors (*L-F*, *L_S-F* and *L_B-F*) are created from the long and short ends of *HML*, *HML_S* and *HML_B*. The investment (*Inv*) spread factor, *CMA* (conservative minus aggressive), is also the average of small stock and big stock spread factors, *CMA_S* and *CMA_B*. The long (*C-F*, *C_S-F*, *C_B-F*) and short (*A-F*, *A_S-F*, *A_B-F*) excess return factors use the long and short ends of *CMA*, *CMA_S*, and *CMA_B*. The profitability (*Prf*) factors are constructed in the same way as the value and investment factors. The profitability factor *RMW_O* for the FF (2015) spread factor model in the first line of the table uses operating profitability. The profitability factors in the remaining lines of the table use cash profitability, indicated with a subscript C. Each line of the table shows (i) the *Mkt*, *Size*, *Val*, *Prf*, and *Inv* factors in a model, (ii) $Sh^2(f)$, the max squared Sharpe ratio for the model's factors, and (iii) the average (Ave), standard deviation (Std Dev), and *t*-statistic for the average of T_f-F , the monthly excess returns on the tangency portfolio implied by $Sh^2(f)$, (iv) the weights for factors in T_f-F , and (v) Short Alloc, the sum of the short positions in the tangency portfolio, expressed as a proportion of the \$1 invested in the tangency portfolio.

Model Factors		$Sh^2(f)$	Monthly Returns (%)			Mkt	Factor Weights In $T_f - F$				Short Alloc
			Ave	Std Dev	t-stat		Size	Val	Prf	Inv	
FF (2015)	$Mkt, SMB, HML, RMW_O, CMA$	0.166	0.33	0.82	10.22	0.10	0.09	0.09	0.43	0.29	-0.90
Spread factors (cash profitability)											
Combined	$Mkt, SMB, HML, RMW_C, CMA$	0.242	0.37	0.76	12.34	0.12	0.11	0.12	0.54	0.11	-0.88
Small	$Mkt, SMB, HML_S, RMW_{CS}, CMA_S$	0.246	0.46	0.93	12.46	0.11	0.12	0.03	0.53	0.20	-0.89
Big	$Mkt, SMB, HML_B, RMW_{CB}, CMA_B$	0.078	0.28	0.99	7.03	0.16	0.10	0.17	0.44	0.13	-0.84
Long excess return factors (cash profitability)											
Combined	$Mkt, S-F, H-F, R_C-F, C-F$	0.195	5.05	11.43	11.09	-14.07	-10.14	2.24	17.64	5.33	-24.21
Small	$Mkt, S-F, H_S-F, R_{CS}-F, C_S-F$	0.160	4.70	11.75	10.03	-0.11	-19.20	3.63	12.84	3.84	-19.31
Big	$Mkt, S-F, H_B-F, R_{CB}-F, C_B-F$	0.089	2.72	9.12	7.50	-14.53	2.45	1.37	9.01	2.70	-14.53
Short excess return factors (cash profitability)											
Combined	$Mkt, S-F, L-F, W_C-F, A-F$	0.185	2.41	5.61	10.80	4.56	4.78	-1.76	-5.49	-1.09	-8.34
Small	$Mkt, S-F, L_S-F, W_{CS}-F, A_S-F$	0.248	3.51	7.06	12.49	-0.29	10.08	0.75	-6.62	-2.92	-9.83
Big	$Mkt, S-F, L_B-F, W_{CB}-F, A_B-F$	0.044	1.16	5.52	5.29	3.74	0.73	1.10	-2.32	-2.23	-4.56

Table 7 – Summary statistics for regression intercepts, July 1963 – December 2015

The 100 LHS portfolios in Panel B are from four 5x5 quintile sorts, first on *ME* and then, independently, on *BE/ME*, *OP*, *CP*, or *Inv*. The 160 LHS portfolios in Panel C are from 2x4x4 sorts first into two *ME* groups and then into quartiles on pairs of *BE/ME*, *Inv*, and either *OP* or *CP*. All sorts use NYSE breakpoints. In the 2x4x4 sorts, the first sort is on *ME* (small or big, using the NYSE median *ME* as the breakpoint) and the other two are conditional on *ME*. Like the factors, the LHS portfolios described above are formed at the end of June of each year *t*, and the variables in the 5x5 and 2x4x4 sorts are the same as those in the 2x3 sorts that produce the factors. The 135 anomaly portfolios in Panel D are from (i) 5x5 sorts on end of June *ME* and market β ; (ii) 5x5 sorts on *ME* and accruals; (iii) 5x5 sorts on end of June *ME* and *Var*, the variance of daily returns; (iv) 5x5 sorts on *ME* and *RVar*, the variance of daily residuals from the FF (1993) three-factor model; and (v) 5x7 sorts on *ME* and negative net share issues (repurchases), zero net share issues, and quintiles of positive net share issues, measured as the implied growth in split-adjusted shares outstanding from June of *t*-1 to June of *t*. (See Appendix.) The first pass *ME* sorts and the second pass anomaly sorts are independent, except in the *Var* and *RVar* sorts. The second pass sorts on *Var* and *RVar* are conditional on *ME* quintile. The 395 LHS portfolios in Panel A are all the LHS portfolios in Panels B (100), C (160), and D (135). The table shows (i) $Sh^2(f)$, the max squared Sharpe ratio for the model's factors, (ii) $Sh^2(a)$, the max squared Sharpe ratio for the intercepts for a set of LHS portfolios, (iii) the *GRS* statistic, (iv) $A|a_i|$, the average absolute intercept, (v) $Aa_i^2/\bar{A}\bar{r}_i^2$, the average squared intercept over the average squared value of $\bar{r}_i$, which is the difference between the average return on LHS portfolio *i* and the average return on the VW Market, and (vi) $As^2(a_i)/Aa_i^2$, the average of the squared sample standard errors of the intercepts over the average squared intercept.

Model	Factors	$Sh^2(f)$	$Sh^2(a)$	<i>GRS</i>	$A a_i $	$\frac{Aa_i^2}{\bar{A}\bar{r}_i^2}$	$\frac{As^2(a_i)}{Aa_i^2}$
Panel A: All LHS portfolios							
Small short excess	<i>Mkt</i> , <i>S-F</i> , <i>L_S-F</i> , <i>W_{CS}-F</i> , <i>A_S-F</i>	0.248	4.735	2.22	0.113	0.206	0.344
Small spread	<i>Mkt</i> , <i>SMB</i> , <i>HML_S</i> , <i>RMW_{CS}</i> , <i>CMA_S</i>	0.246	4.756	2.24	0.102	0.184	0.393
Combined spread	<i>Mkt</i> , <i>SMB</i> , <i>HML</i> , <i>RMW_C</i> , <i>CMA</i>	0.242	4.816	2.27	0.103	0.178	0.399
Com. long excess	<i>Mkt</i> , <i>S-F</i> , <i>H-F</i> , <i>R_C-F</i> , <i>C-F</i>	0.195	4.929	2.42	0.123	0.276	0.251
Com. short excess	<i>Mkt</i> , <i>S-F</i> , <i>L-F</i> , <i>W_C-F</i> , <i>A-F</i>	0.185	4.892	2.42	0.115	0.239	0.287
FF (2015)	<i>Mkt</i> , <i>SMB</i> , <i>HML</i> , <i>RMW_O</i> , <i>CMA</i>	0.166	4.873	2.45	0.106	0.197	0.333
Small long excess	<i>Mkt</i> , <i>S-F</i> , <i>H_S-F</i> , <i>R_{CS}-F</i> , <i>C_S-F</i>	0.160	4.901	2.48	0.134	0.306	0.220
Big long excess	<i>Mkt</i> , <i>S-F</i> , <i>H_B-F</i> , <i>R_{CB}-F</i> , <i>C_B-F</i>	0.089	4.989	2.69	0.158	0.474	0.136
Big spread	<i>Mkt</i> , <i>SMB</i> , <i>HML_B</i> , <i>RMW_{CB}</i> , <i>CMA_B</i>	0.078	4.962	2.70	0.179	0.564	0.113
Big short excess	<i>Mkt</i> , <i>S-F</i> , <i>L_B-F</i> , <i>W_{CB}-F</i> , <i>A_B-F</i>	0.044	5.094	2.86	0.209	0.754	0.083
Correlation with $Sh^2(f)$		1.000	-0.931	-0.995	-0.924	-0.926	0.950

Table 7 (continued)

Model	Factors	$Sh^2(f)$	$Sh^2(a)$	GRS	$A a_i $	$\frac{Aa_i^2}{A\bar{r}_i^2}$	$\frac{As^2(a_i)}{Aa_i^2}$
Panel B: 5x5 portfolios							
Small short excess	$Mkt, S-F, L_S-F, W_{CS}-F, A_S-F$	0.248	0.498	2.12	0.103	0.175	0.303
Small spread	$Mkt, SMB, HML_S, RMW_{CS}, CMA_S$	0.246	0.509	2.16	0.075	0.100	0.545
Combined spread	$Mkt, SMB, HML, RMW_C, CMA$	0.242	0.508	2.17	0.083	0.130	0.412
Com. long excess	$Mkt, S-F, H-F, R_C-F, C-F$	0.195	0.547	2.43	0.083	0.157	0.329
Com. short excess	$Mkt, S-F, L-F, W_C-F, A-F$	0.185	0.571	2.55	0.094	0.185	0.281
FF (2015)	$Mkt, SMB, HML, RMW_O, CMA$	0.166	0.590	2.68	0.078	0.128	0.384
Small long excess	$Mkt, S-F, H_S-F, R_{CS}-F, C_S-F$	0.160	0.595	2.72	0.095	0.179	0.285
Big long excess	$Mkt, S-F, H_B-F, R_{CB}-F, C_B-F$	0.089	0.654	3.18	0.120	0.342	0.142
Big spread	$Mkt, SMB, HML_B, RMW_{CB}, CMA_B$	0.078	0.667	3.27	0.150	0.436	0.110
Big short excess	$Mkt, S-F, L_B-F, W_{CB}-F, A_B-F$	0.044	0.707	3.58	0.171	0.575	0.084
Correlation with $Sh^2(f)$		1.000	-0.997	-0.998	-0.850	-0.901	0.882
Panel C: 2x4x4 portfolios							
Small short excess	$Mkt, S-F, L_S-F, W_{CS}-F, A_S-F$	0.248	0.666	1.57	0.112	0.165	0.395
Small spread	$Mkt, SMB, HML_S, RMW_{CS}, CMA_S$	0.246	0.661	1.55	0.100	0.134	0.490
Combined spread	$Mkt, SMB, HML, RMW_C, CMA$	0.242	0.674	1.59	0.091	0.116	0.541
Com. long excess	$Mkt, S-F, H-F, R_C-F, C-F$	0.195	0.715	1.75	0.110	0.183	0.336
Com. short excess	$Mkt, S-F, L-F, W_C-F, A-F$	0.185	0.728	1.80	0.115	0.196	0.312
FF (2015)	$Mkt, SMB, HML, RMW_O, CMA$	0.166	0.780	1.96	0.103	0.152	0.382
Small long excess	$Mkt, S-F, H_S-F, R_{CS}-F, C_S-F$	0.160	0.746	1.89	0.132	0.235	0.261
Big long excess	$Mkt, S-F, H_B-F, R_{CB}-F, C_B-F$	0.089	0.822	2.21	0.156	0.401	0.144
Big spread	$Mkt, SMB, HML_B, RMW_{CB}, CMA_B$	0.078	0.834	2.27	0.175	0.488	0.117
Big short excess	$Mkt, S-F, L_B-F, W_{CB}-F, A_B-F$	0.044	0.858	2.41	0.223	0.719	0.080
Correlation with $Sh^2(f)$		1.000	-0.987	-0.996	-0.912	-0.915	0.948
Panel D: Anomaly portfolios							
Small short excess	$Mkt, S-F, L_S-F, W_{CS}-F, A_S-F$	0.248	0.886	2.60	0.122	0.292	0.315
Small spread	$Mkt, SMB, HML_S, RMW_{CS}, CMA_S$	0.246	0.930	2.73	0.125	0.315	0.298
Combined spread	$Mkt, SMB, HML, RMW_C, CMA$	0.242	0.878	2.59	0.131	0.307	0.313
Com. long excess	$Mkt, S-F, H-F, R_C-F, C-F$	0.195	0.972	2.98	0.169	0.498	0.186
Com. short excess	$Mkt, S-F, L-F, W_C-F, A-F$	0.185	0.859	2.65	0.131	0.340	0.268
FF (2015)	$Mkt, SMB, HML, RMW_O, CMA$	0.166	0.932	2.93	0.130	0.313	0.283
Small long excess	$Mkt, S-F, H_S-F, R_{CS}-F, C_S-F$	0.160	0.985	3.11	0.165	0.498	0.174
Big long excess	$Mkt, S-F, H_B-F, R_{CB}-F, C_B-F$	0.089	0.987	3.32	0.190	0.674	0.126
Big spread	$Mkt, SMB, HML_B, RMW_{CB}, CMA_B$	0.078	0.961	3.26	0.206	0.765	0.110
Big short excess	$Mkt, S-F, L_B-F, W_{CB}-F, A_B-F$	0.044	0.975	3.42	0.221	0.926	0.087
Correlation with $Sh^2(f)$		1.000	-0.653	-0.935	-0.916	-0.927	0.920